DynoX thrust stand

RTS Pro 20

User manual · 20 kgf thrust and torque test stand



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Designed and manufactured by REUDE Technologies Private Limited, Chennai, India. Read the safety instructions in Section 3 before assembling or operating this equipment.

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1. About this manual

This manual explains how to assemble, wire, calibrate, operate and maintain the RTS Pro 20. It is written for technicians, engineers and lab staff with basic mechanical and electrical skills. Keep it with the stand and pass it on with the equipment.

How to read the safety notes

WARNING A hazard that can cause serious injury. Follow the instruction exactly.

CAUTION A risk of damage to the stand, the sensors or the measurement.

NOTE Useful information that makes the job easier or the result better.

Interactive 3D model

Every component in this manual is numbered the same way as in the DynoX 3D viewer. Open the viewer to rotate the stand, see the exploded sub-assemblies, and lay every component out on the table.

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2. Product overview

The RTS Pro 20 is a floor-standing thrust stand for heavy-lift drone propulsion up to 20 kgf of thrust. A braced twin-pillar tower holds the motor about 1.1 m above the floor, clear of ground effect. Factory-calibrated load cells measure thrust and torque together, while the ROTRIX DAQ unit records RPM, voltage, current, vibration and temperature in sync and streams them to the ROTRIX software.

Specifications

Item	Specification
Rated thrust	0 to 20 kgf
Measured parameters	Thrust, torque, RPM, voltage, current, vibration and temperature
Force sensing	Load-cell sensor block, ±0.5% of full scale
Speed sensing	Optical IR sensor, 0 to 25,000 RPM, ±1%
Electrical sensing	Inline current sensor module on the right pillar (5 to 100 A); voltage 3S to 12S measured by the DAQ
Vibration and temperature	Vibration ±16 g; temperature -55 to 125 °C
Data acquisition	ROTRIX DAQ unit, STM32 controller, signal conditioning, filtering and protection; USB link
Software	ROTRIX test control and logging; ROTRIDASH analytics (optional)
Frame	40-series aluminium extrusions with orange slot covers (REUDE standard)
Footprint	About 845 × 1,100 mm including stabiliser arms
Thrust axis height	About 1,080 mm above the floor
Reference propeller	About 1,030 mm (40 inch) two-blade

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Main components

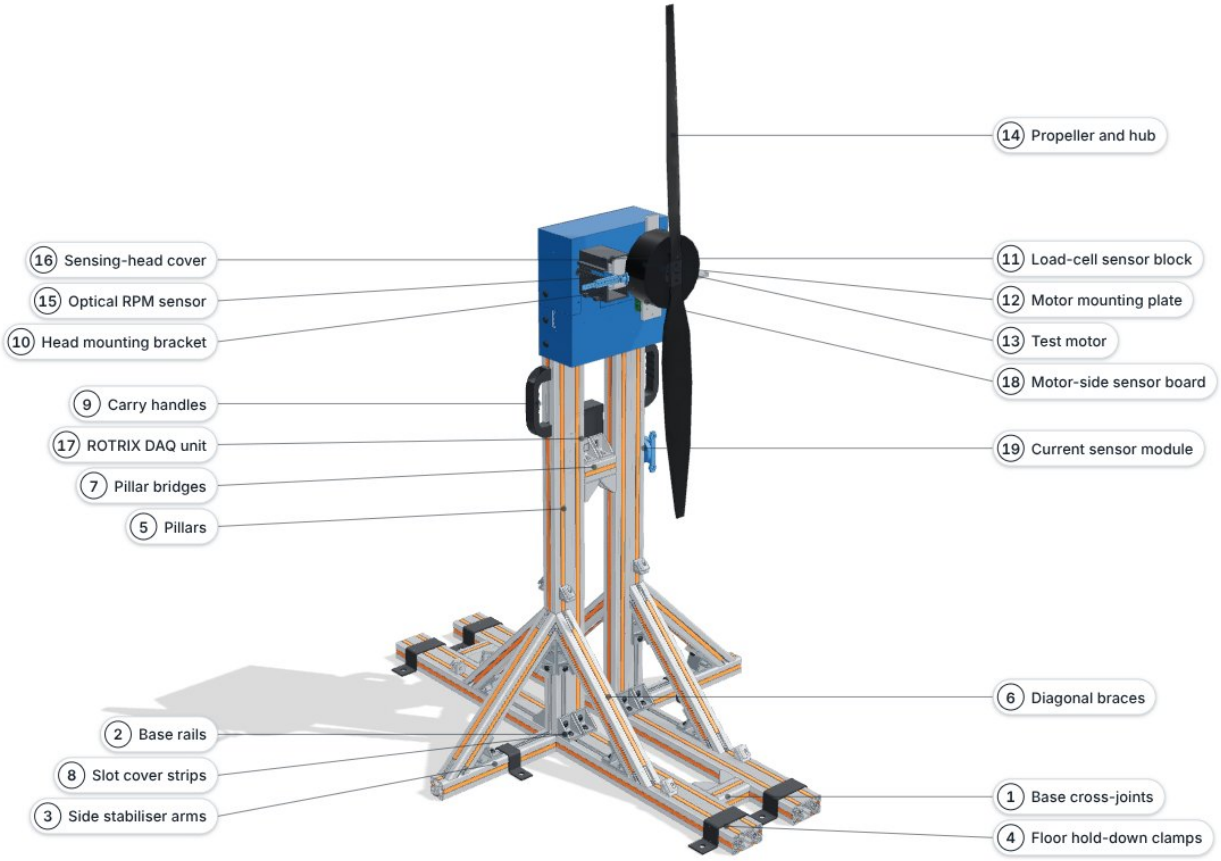


Figure 2.1 RTS Pro 20 assembled, with the main components numbered.

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Exploded view

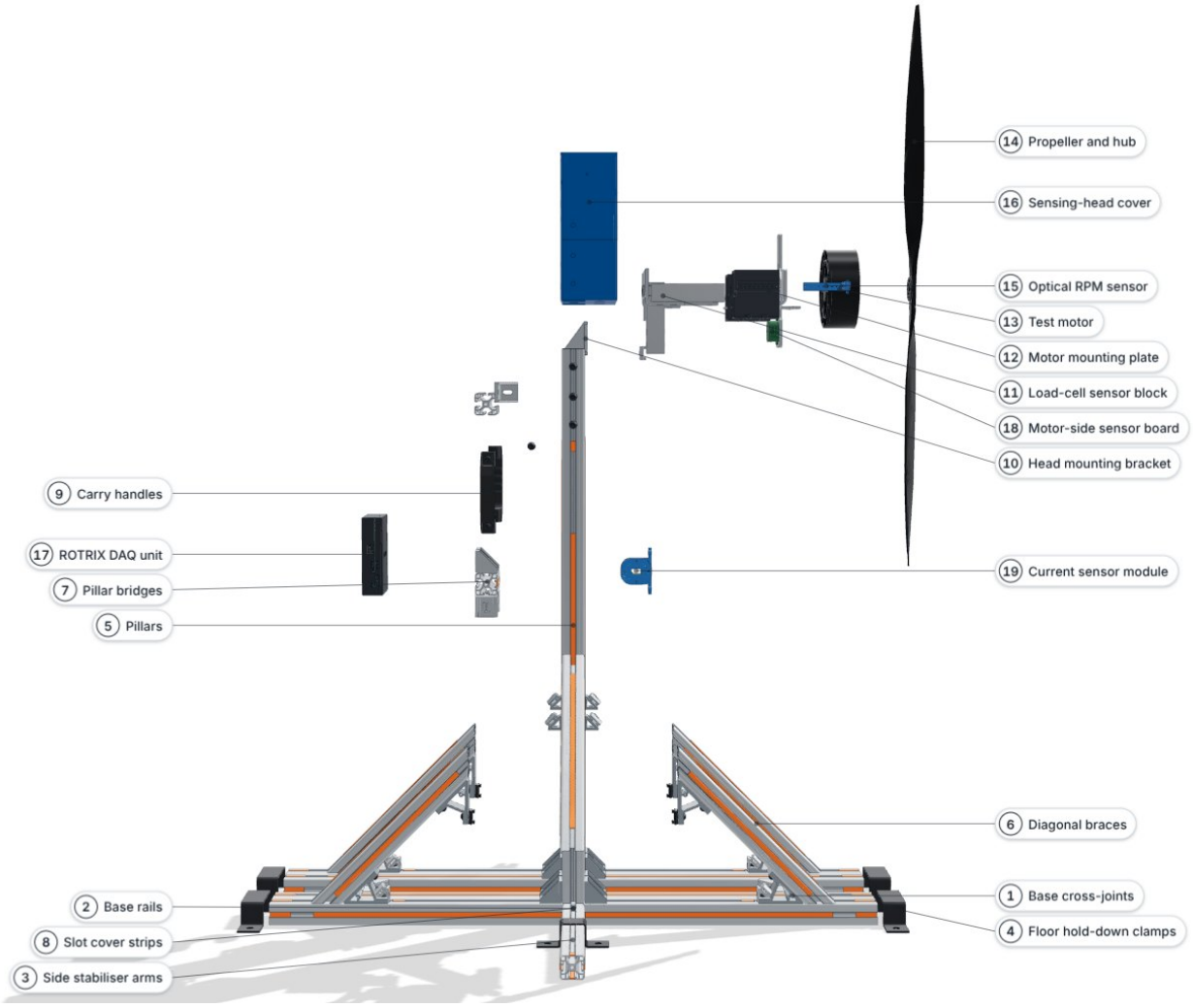


Figure 2.2 Exploded side view. Each sub-assembly moves as one block, in the order it is assembled.

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3. Safety instructions

WARNING A spinning propeller can cause severe injury. Never approach the stand while a battery is connected, and keep everyone out of the propeller plane during a test.

Assembly team

At least three people are required: two to hold the pillars and one to fit brackets and bolts.

People and protection

- Read this manual completely before assembling or operating the stand.
- Wear safety glasses and gloves during assembly. Wear safety glasses and hearing protection during every test run.
- Tie back long hair and remove loose clothing, lanyards and jewellery before approaching a running motor.
- Never reach over, under or past a propeller while a battery is connected, even if the motor is stopped.

Test area

- Place the stand on a level, rigid floor or bench and anchor it with the supplied clamps before any powered test.
- Mark an exclusion zone around the stand. Nobody may stand in the plane of the propeller disc while a battery is connected.
- Keep the area clear of loose paper, tools, cable offcuts and anything that the propeller wash can lift.
- Test indoors or under cover. The sensors and the DAQ unit are not weatherproof: keep them away from rain, condensation and dust.

Electrical and battery

- Connect the battery last and disconnect it first. Keep the throttle at zero and the ESC disarmed while connecting.
- Use cables, connectors and fuses rated above the maximum test current. Check insulation before every session.
- Fit an emergency-stop or kill switch in the power line, within reach of the operator, and test it before each session.
- Charge, store and transport lithium batteries in fire-safe bags or containers, and never leave a charging battery unattended.
- When the operator works at a distance, only the digital data link and the emergency-stop loop should run to the operator station. Do not extend the analogue sensor cables.

Operation

- Never exceed the rated thrust of the stand. Stop the test if a reading approaches the limit.
- Before each run, check that every bolt is tight, the propeller is undamaged and balanced, and its hub bolts are secure.
- Start every new motor and propeller combination at low throttle and increase in steps while watching vibration and temperature.
- Stop immediately on unusual noise, rising vibration, smoke, or any reading that does not make sense.

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- Do not lean on, lift or carry the stand by the sensing head. Loads on the head can permanently damage the load cells.

CAUTION The rated thrust of this stand is 20 kgf. Loads above the rating can permanently damage the load cells and void the calibration.

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4. Package contents

Check every item against this list before you start. If anything is missing or damaged, contact REUDE support before assembly. Do not start with incomplete or faulty parts.

No.	Component	Code	Size	Material	Qty
1	**Base cross-joints**	SUB-ASSY-001 · Profile B with 90° brackets	40 × 40 × 85 mm profile, two 90° corner brackets each	Anodised aluminium extrusion, die-cast aluminium brackets	2
2	**Base rails**	SKU-EX40A · Profile A	40 × 80 mm section, 1,100 mm long	Anodised aluminium extrusion	2
3	**Side stabiliser arms**	40 × 40 profile	40 × 40 × 300 mm	Anodised aluminium extrusion	2
4	**Floor hold-down clamps**	Anchor clamps	4 × 165 × 40 mm on the rail ends, 2 × 125 × 30 mm on the arm ends	Steel clamp plate	6
5	**Pillars**	SKU-EX40A · Profile A	40 × 80 × 1,000 mm	Anodised aluminium extrusion	2
6	**Diagonal braces**	40 × 40 profile at 45°	40 × 40 mm profile, about 400 mm long, 45° brackets at each end	Anodised aluminium extrusion, die-cast aluminium brackets	6
7	**Pillar bridges**	SKU-EX40D-0085 · Profile D	40 × 40 × 85 mm	Anodised aluminium extrusion	2
8	**Slot cover strips**	Orange T-slot covers · REUDE standard	9 mm wide, cut to length for each exposed slot	Flexible PVC, REUDE orange	1 set
9	**Carry handles**	Pillar grips	About 145 mm long	Moulded handle with steel screws	2
10	**Head mounting bracket**	Gusseted mount	80 × 60 mm base with side gussets	Aluminium	1
11	**Load-cell sensor block**	Thrust and torque sensing	Carrier 200 × 90 × 70 mm, load cells 32 × 32 × 130 mm	Aluminium carrier, precision strain-gauge load cells	1 block, 3 load cells
12	**Motor mounting plate**	Motor interface	200 × 207 mm plate with two clamp housings	Aluminium plate	1
13	**Test motor**	Brushless outrunner (reference)	Ø145 × 56 mm	Brushless DC motor	1
14	**Propeller and hub**	Two-blade, about 40 inch	About 1,030 mm diameter	Carbon-fibre composite blades, metal hub adapter	2 blades + hub
15	**Optical RPM sensor**	IR blade counter	105 mm arm with IR sensor head	3D-printed arm, optical IR sensor	1
16	**Sensing-head cover**	Protective housing	251 × 261 × 98 mm	Powder-coated sheet metal	1
17	**ROTRIX DAQ unit**	STM32-based data acquisition	88 × 143 × 38 mm	Custom PCB with signal conditioning, in an enclosure	1

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No.	Component	Code	Size	Material	Qty
18	**Motor-side sensor board**	Vibration and temperature	52 × 50 mm PCB	FR-4 circuit board	1
19	**Current sensor module**	Inline current sensor	24 × 90 × 58 mm	3D-printed case, current sensor board	1

Brackets and fixings

No.	Fixing	Code	Size	Material	Qty
20	**90° corner brackets**	SKU-CB90L-0035	40 × 40 × 35 mm	Die-cast aluminium	19
21	**45° brace brackets**	Diagonal end brackets	35 × 58 × 80 mm	Die-cast aluminium	12
22	**45° angle bracket sets**	SKU-AB45L-0005	About 21 × 24 × 24 mm	Die-cast aluminium with steel bolt and washer	8 sets
23	**T-slot sliding nuts**	SKU-SN90T-0019	19.5 × 19.5 × 9.5 mm	Zinc-plated steel	42
24	**M8 button-head bolts**	SKU-BB08M-0010	M8 × 10 mm	Steel, black oxide	49

Component importance

Critical parts are in the load path or the measurement chain. **Important** parts set the stiffness and alignment of the frame. **Standard** parts are for convenience and protection.

No.	Component	Importance	Why it matters
1	Base cross-joints	**Important**	They keep the rails parallel and square. A loose joint lets the frame rack, which tilts the thrust line and makes readings drift.
2	Base rails	**Critical**	Their length sets the footprint. Up to 20 kgf of thrust acts about one metre above the floor, and the rails are what stop that force from tipping the stand.
3	Side stabiliser arms	**Important**	Motor torque tries to roll the stand sideways. The arms give it a wide stance so it stays planted at full power.
4	Floor hold-down clamps	**Critical**	Thrust tries to lift and slide the stand. Anchoring it keeps the measurement clean and keeps the test area safe.
5	Pillars	**Critical**	They hold the propeller axis about 1.1 m above the floor, away from ground effect, and carry all the thrust down into the base. Pillars that are out of square tilt the thrust line and add error.
6	Diagonal braces	**Critical**	Triangles are the stiffest shape in a structure. The braces damp motor vibration and stop fore-aft and sideways sway, so the load cells read thrust, not wobble.
7	Pillar bridges	**Important**	They stop the pillars twisting or spreading under oscillating load. The upper bridge also carries the sensing head.
8	Slot cover strips	**Standard**	They keep dust, swarf and small parts out of the slots, cover the sharp slot edges, hold cables neatly, and give the stand its signature orange accent.
9	Carry handles	**Standard**	Makes the stand easier and safer to move. Handles are outside the load path and do not affect measurement.
10	Head mounting bracket	**Critical**	It is the only link between the sensors and the frame. Any play here appears as noise in the thrust and torque readings.

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No.	Component	Importance	Why it matters
11	Load-cell sensor block	Critical	The load cells turn up to 20 kgf of thrust, and the motor’s reaction torque, into electrical signals at ±0.5% of full scale. Every result the stand produces depends on them.
12	Motor mounting plate	Critical	Its large, stiff face keeps the motor square to the measurement axis, so all of the thrust is measured and none is lost to flex.
13	Test motor	Important	This is the unit under test. Any motor within the 20 kgf rating can be fitted to the mounting plate.
14	Propeller and hub	Critical	At full power the blade tips move very fast. Good balance, tight hub bolts and a clear zone around the disc are essential for safety and clean vibration data.
15	Optical RPM sensor	Important	Speed (0 to 25,000 RPM, ±1%) is needed for efficiency figures and thrust-versus-RPM curves.
16	Sensing-head cover	Standard	It shields the load cells and wiring from dust, knocks and propeller wash, which keeps readings stable.
17	ROTRIX DAQ unit	Critical	It samples thrust, torque, RPM, voltage, current, vibration and temperature in sync, filters and protects the signals, and sends them to your PC over USB.
18	Motor-side sensor board	Important	Placed close to the motor, it catches imbalance and overheating early, before they damage the motor or the stand.
19	Current sensor module	Important	Current, with voltage, gives electrical power, so it is needed for every efficiency figure.

All components laid out

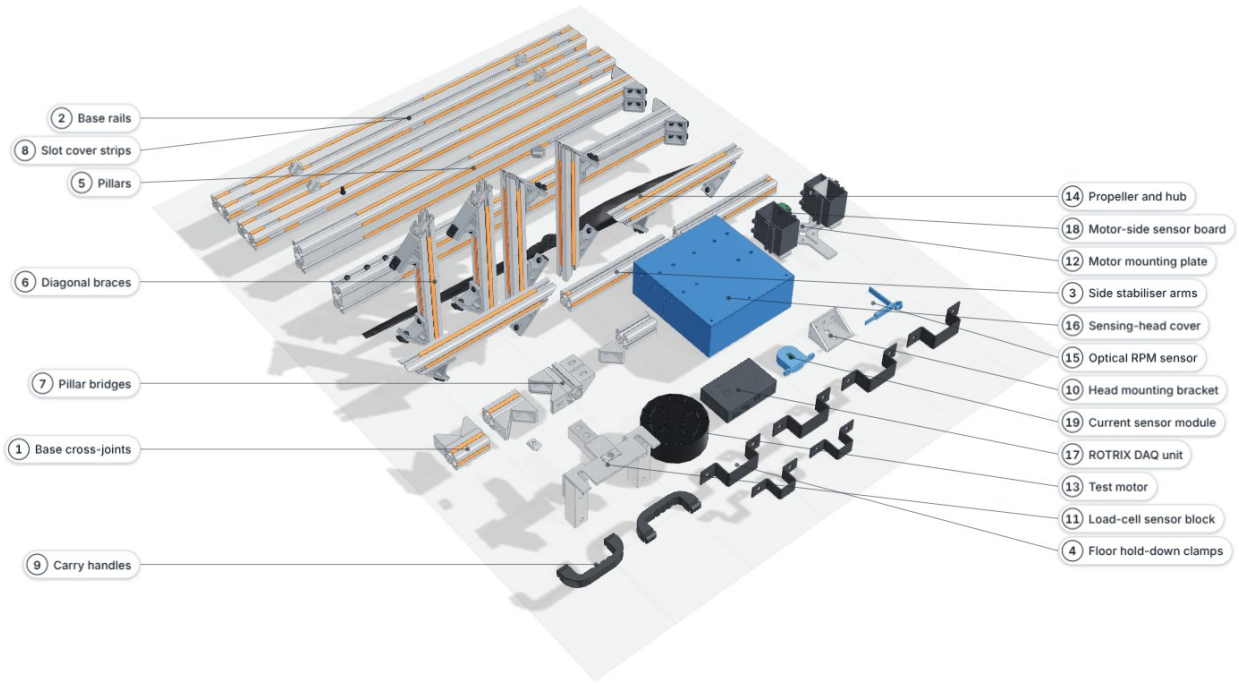


Figure 4.1 The components laid out on the table, as shipped.

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5. Tools and tightening torques

Tools you will need

- Hex keys 4, 5 and 6 mm
- Spanner or socket set
- Torque wrench
- Spirit level, try square and tape measure
- Cable ties and a side cutter
- Multimeter and clamp meter
- Hand tachometer (for the RPM check)
- Calibration weights (for example 2, 5 and 10 kg) with cord and a low-friction pulley
- Masonry drill and M12 anchors for concrete floors

Recommended tightening torques

Fastener	Where used	Typical tightening
M3 screws	Printed parts, sensor boards, DAQ enclosure	Hand tight, about 0.5 to 1 Nm. Do not crack the printed parts.
M6 bolts	Motor to mounting plate, plate to bracket	About 8 Nm, with thread-locker on motor bolts
M8 button-head bolts	Brackets into T-slot nuts	About 12 to 15 Nm into steel T-nuts; snug only (about 4 Nm) into printed T-nuts. Tighten evenly around each joint.

NOTE Tighten in two passes: first snug every bolt in a joint, then bring each to the final torque. This keeps the frame square.

CAUTION Always load the T-slot nuts into a profile before a bracket, end cap or another profile closes that slot. A missing nut means taking the joint apart again.

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6. Assembly, step by step

Work through the steps in order. In each figure the parts added in that step are highlighted in orange; parts fitted in earlier steps are shown in their final position.

Step 1. Base cross-joints (SUB-ASSY-001)



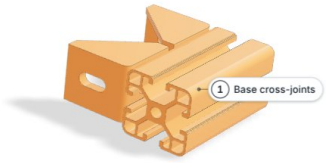


Figure 6.1 Base cross-joints (SUB-ASSY-001).

1. Take one short profile, two 90° corner brackets, two T-nuts and two M8 bolts.
2. Slide a T-nut into the top slot on each side of the profile, roughly where the brackets will sit.
3. Hold a bracket flush against each side, line its hole up with the nut and fit the M8 bolt.
4. Tighten until snug, check the brackets sit flush, then build a second identical joint.

CAUTION Do not over-tighten. Excess torque can strip the T-nut or distort the bracket.

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Step 2. Base frame (SUB-ASSY-002)

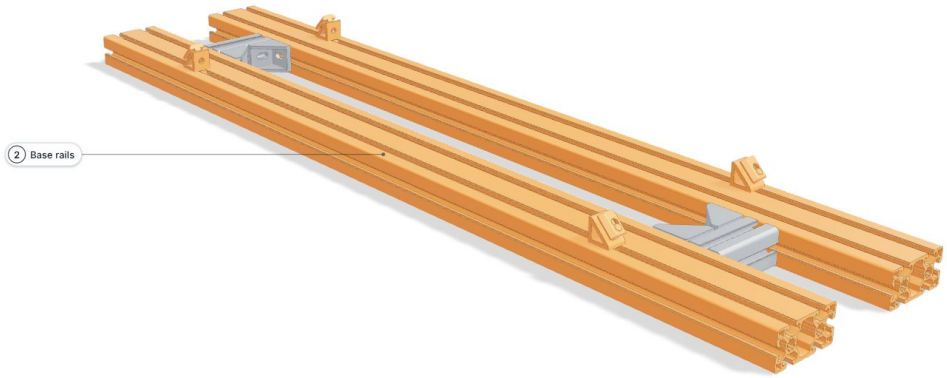


Figure 6.2 Base frame (SUB-ASSY-002).

1. Insert T-nuts into both base rails for every bracket that will land on them: the cross-joints, the pillar feet, the braces, the stabiliser arms and the clamps.
2. Place a cross-joint between the rails at each end, symmetric and flush with the central slot.
3. Bolt the cross-joint brackets into the rail nuts.
4. Measure both diagonals of the frame. They should match within 2 mm. Adjust, then tighten.

CAUTION Loading every T-nut now saves dismantling later. The T-nuts cannot be added once a slot is closed.

Step 3. Stabiliser arms and floor clamps

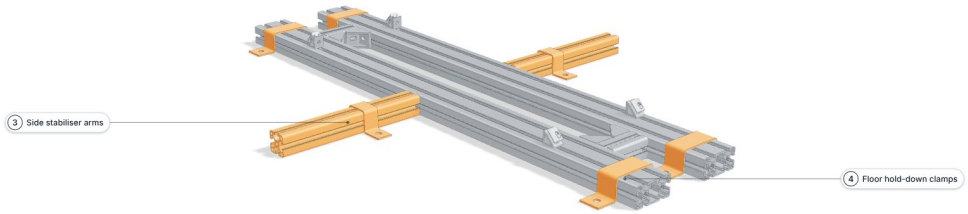


Figure 6.3 Stabiliser arms and floor clamps.

1. Bolt a 300 mm stabiliser arm square to the outer face of each base rail, in line with the pillar positions.
2. Move the frame to its final position on a level floor.
3. Place the four rail-end clamps and the two arm-end clamps. Mark the anchor holes, but leave the clamps loose until Step 6.

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Step 4. Pillars (SUB-ASSY-003)

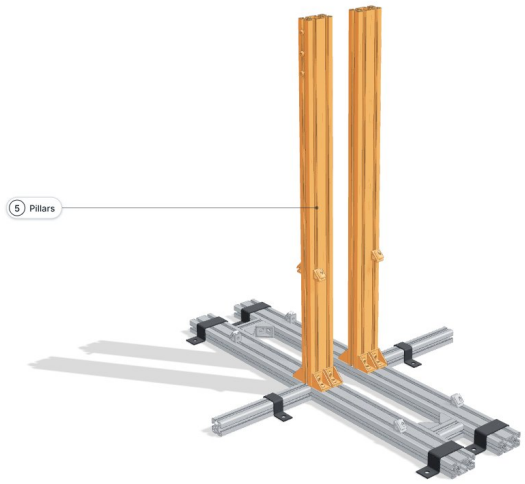


Figure 6.4 Pillars (SUB-ASSY-003).

1. Insert T-nuts into both pillars for the braces, the bridges, the handles and the DAQ unit.
2. With two people holding, stand both 1,000 mm pillars upright at the centre of the base.
3. Fit four 90° corner brackets around each pillar foot (eight in total) with M8 bolts.
4. Check that the pillars are vertical on two faces and parallel to each other, then tighten.

CAUTION A pillar can topple before its brackets are tight. Keep two people holding the pillars until all eight brackets are bolted.

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Step 5. Diagonal braces (SUB-ASSY-004)



Figure 6.5 Diagonal braces (SUB-ASSY-004).

1. Fit a 45° end bracket at each end of every brace.
2. Offer each front and rear brace in at 45° between the base rail and the pillar, and bolt both ends.
3. Fit the two side braces between the stabiliser arms and the pillars in the same way.
4. Tighten all brace bolts evenly, then recheck that the pillars are vertical.

Step 6. Pillar bridges, handles and anchoring (SUB-ASSY-005)



Figure 6.6 Pillar bridges, handles and anchoring (SUB-ASSY-005).

1. Place the lower and upper 85 mm bridges between the pillars at their marked heights, level and centred.
2. Fix each bridge with 90° corner brackets and M8 bolts.
3. Screw a carry handle to the outer slot of each pillar.
4. Tighten the whole frame to the recommended torque, then drill, anchor and tighten all six floor clamps.

CAUTION Anchor the clamps only after the frame is fully squared and tightened.

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Step 7. Sensing head (SUB-ASSY-006)



Figure 6.7 Sensing head (SUB-ASSY-006).

1. Bolt the head mounting bracket onto the upper bridge and check that it is square to the pillars.
2. Bolt the load-cell sensor block onto the bracket with the motor face pointing forward. Do not dismantle the block.
3. Fit the motor mounting plate, with its sensor board, to the front of the sensor block.

CAUTION The load-cell block is calibrated at the factory as one unit. Taking it apart voids the calibration.

Step 8. Motor, propeller and RPM sensor

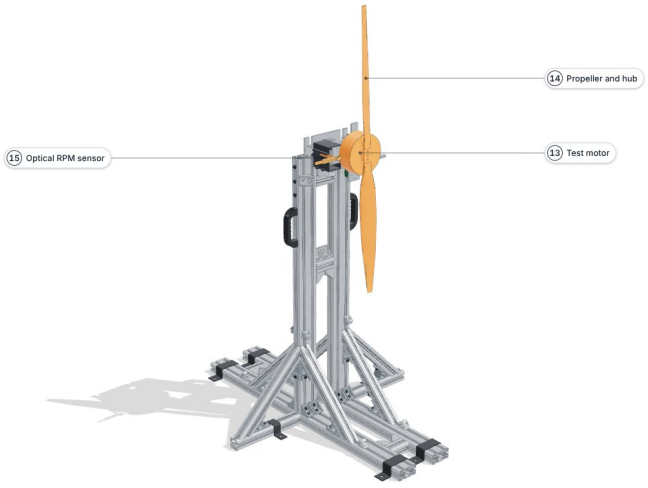


Figure 6.8 Motor, propeller and RPM sensor.

1. Bolt the motor to the mounting plate with M6 bolts and thread-locker, and check that the shaft is centred.
2. Fix the RPM sensor arm on the propeller side and aim the sensor at the blade path.
3. Fit the propeller hub and blades last, after wiring is complete and with the battery disconnected.

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Step 9. DAQ unit, current sensor and sensing-head cover



Figure 6.9 DAQ unit, current sensor and sensing-head cover.

1. Mount the DAQ unit on the rear face of the pillars at mid height.
2. Screw the current sensor case to the outer slot of the right pillar, below the handle.
3. Complete the wiring in Section 8, then fit the sensing-head cover without trapping any cable.

Step 10. Orange slot cover strips (REUDE standard)



Figure 6.10 Orange slot cover strips (REUDE standard).

1. Check that every bolt in the frame is at its final torque and that all cables are routed.
2. Press an orange cover strip into each open T-slot, starting at one end and working along by hand.
3. Cut the strip flush at each bracket and at the profile ends with a sharp knife.
4. Where a cable runs inside a slot, lay it in the slot first and press the cover over it.

CAUTION Fit the covers last. They are easy to lift out later if you need to add a T-nut or re-route a cable.

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Sensing head, exploded

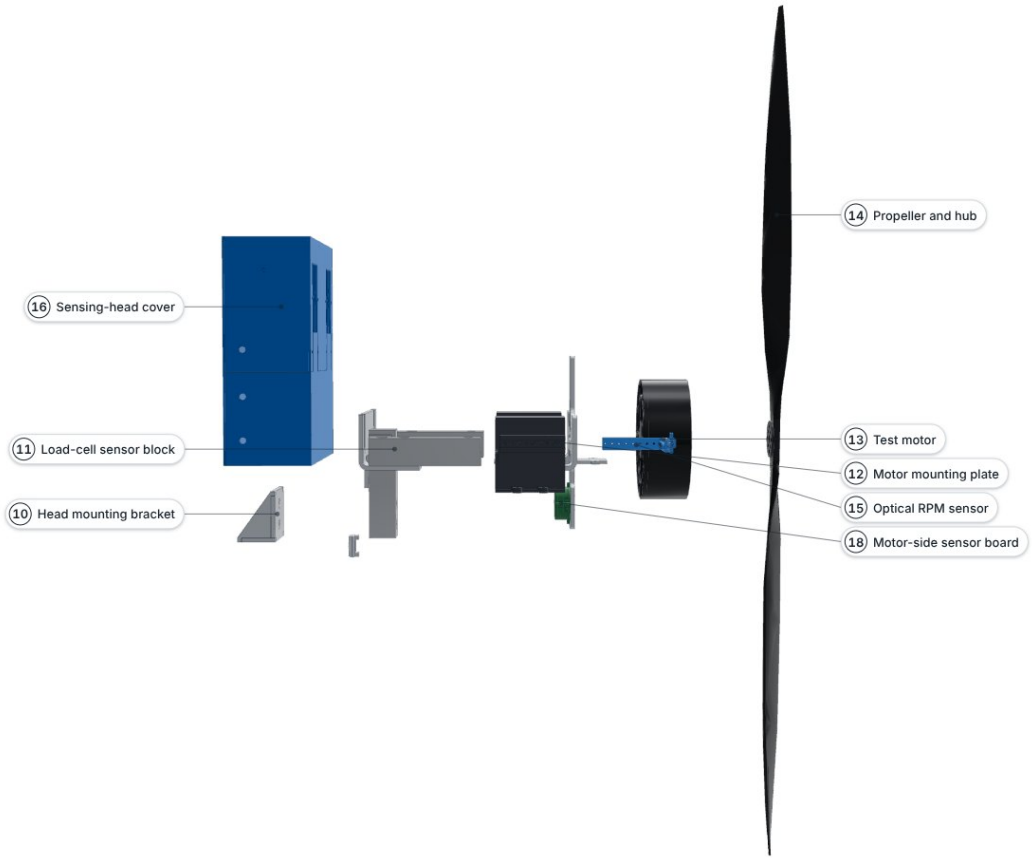


Figure 6.11 Exploded side view of the sensing head, motor and propeller.

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7. DAQ and electronics assembly

The ROTRIX DAQ unit mounts on the rear face of the pillars, close to the sensing head and away from the propeller. It reads every sensor in sync and sends the data to the PC over USB.

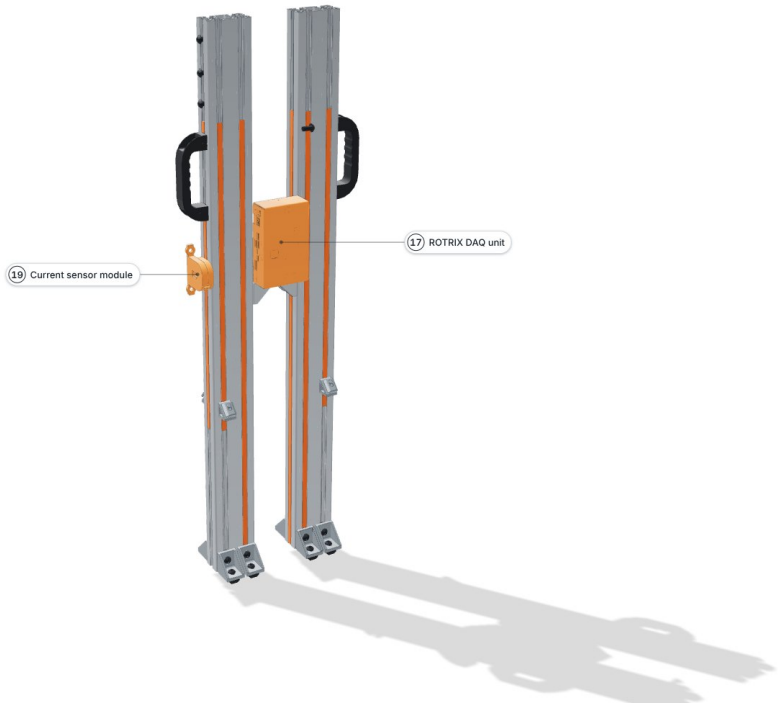


Figure 7.1 ROTRIX DAQ unit on the rear of the pillars, and the current sensor module on the right pillar below the handle.

Mounting and connecting

1. Slide two T-nuts into the rear slot of each pillar at mid height.
2. Bolt the DAQ unit to the pillars with its connector side facing up.
3. Connect the load-cell sensor block, RPM sensor and motor-side sensor board to the matching DAQ inputs.
4. Fit the current sensor board in its case on the right pillar, close the cover and run the battery-to-ESC lead through it, arrow pointing from the battery to the ESC.
5. Connect the current sensor signal lead and the voltage sensing lead to the DAQ unit.
6. Connect the DAQ unit to the PC with the USB cable and power it up.

What the DAQ unit measures

- Thrust and torque from the load cells, through precision bridge amplifiers.
- Motor speed from the optical RPM sensor.
- Battery voltage and current in the power line to the ESC.
- Vibration and temperature from the sensor boards.
- All channels are time-stamped together and streamed to ROTRIX for live display, logging and reports.

CAUTION Connect and disconnect sensor cables only with the DAQ unit powered off and the battery disconnected.

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8. Wire routing to the sensors

Good cable routing matters as much as good mechanics. A cable that is pulled tight across the sensing head carries part of the load, and the load cells then measure the cable instead of the motor. Follow the routes and rules below.



Figure 8.1 Recommended cable routes. Signal cables and power leads run on separate sides of the frame.

	Cable
	Load-cell cables
	RPM sensor
	Vibration and temperature sensor
	Current sensor
	Battery power (through the current sensor)
	USB to PC

Routing table

Cable	From	Route	To
Load-cell cables	Sensor block	Service loop at the head, then down the rear face of the left pillar, tied every 150 mm	DAQ load-cell inputs
RPM sensor	RPM arm	Back along the head, down the left pillar	DAQ RPM input
Vibration and temperature	Motor-side sensor board	Service loop under the head, down the left pillar	DAQ sensor inputs
Current sensor signal	Module on the right pillar	Short lead round the right pillar to the DAQ unit	DAQ current input

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Cable	From	Route	To
Power leads	Battery	Up the right pillar, through the current sensor, past the handle to the ESC beside the head	ESC, then motor
USB	DAQ unit	Down the left pillar and along the floor to the operator station	PC

Routing rules

1. **Leave a service loop.** Where a cable leaves the sensing head, form a loose loop about 50 mm across, so the cable carries no load when the head deflects.
2. **Tie to the fixed frame only.** Fix cables to the frame with ties about every 150 mm. Never tie a cable to the motor plate, the load cells or the free end of the sensing head.
3. **Keep signal and power apart.** Run sensor cables and power leads on opposite sides of the frame. Where they must cross, cross at a right angle.
4. **Stay out of the propeller disc.** Route every cable behind the motor plate and away from the plane of the propeller.
5. **Relieve strain at the DAQ.** Leave a little slack at each connector and label both ends of every cable.
6. **Screened load-cell cables.** Connect the screen at the DAQ end only.
7. **Remote operation.** If the operator sits away from the stand, extend only the USB data link and the emergency-stop loop. Never extend the analogue sensor cables.
8. **Current sensor direction.** Fit the current sensor with its arrow pointing from the battery towards the ESC.
9. **Use the slots.** On the fixed frame, lay cables inside the T-slots and press the orange slot covers over them. This keeps cables tidy, protected and clear of the propeller.

CAUTION After any change to the wiring, zero the load cells again and check that the zero reading stays steady for one minute before testing.

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9. Commissioning and calibration

Before first power-up

- All bolts tightened to the recommended torque, and the base anchored.
- Motor fitted and centred; propeller not yet fitted.
- All cables routed with service loops, and the emergency stop tested.

Connect and check the sensors

1. Connect the DAQ unit to the PC over USB and open ROTRIX.
2. Confirm that every sensor is detected and shows a sensible live value.
3. Allow 10 minutes for the load cells to warm up, then zero (tare) thrust and torque with the motor fitted and stationary.

Check thrust and torque with known loads

1. Thrust: tie a cord to the motor hub, run it over a low-friction pulley in line with the thrust axis, and hang known weights (2, 5 and 10 kg). Compare each reading with the weight.
2. Torque: fit a light lever arm of known length to the motor plate and hang a known weight from its end. The expected torque is weight × 9.81 × arm length (N·m, with the weight in kg and the length in m).
3. If any reading is outside the stated accuracy, run the calibration routine in ROTRIX and repeat the check.

Check RPM, voltage and current

- Compare the RPM reading with a hand tachometer at two or three throttle settings.
- Compare voltage and current with a multimeter and a clamp meter.

First run

1. Fit and balance the propeller, and clear the exclusion zone.
2. Connect the battery last. Arm the ESC from the operator station.
3. Run at about 10% throttle for 30 seconds, watching vibration, temperature and noise.
4. Increase in steps of 10% only while every reading stays stable.

10. Operating the stand

- Do not exceed 20 kgf of thrust. Set a thrust limit alarm in ROTRIX below the rating.
- Run a short warm-up before each test series and zero the load cells with the motor stopped.
- Use ROTRIX to run automated throttle steps and ramps, and ROTRIDASH to compare runs and produce PDF performance reports.
- Allow the motor and ESC to cool between long runs, and watch the temperature channels.
- After testing, disarm the ESC, disconnect the battery, then stop logging and save the data.

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11. Maintenance

When	Check
Before every session	Bolts on the sensing head and motor; propeller condition and balance; cable routing and service loops; clamps anchored; emergency stop works
Weekly in regular use	All frame bolts at the recommended torque; sensors clean; no cracks in brackets or printed parts
Every month	Zero and span check of the thrust and torque readings with known weights; RPM check against a hand tachometer
Every 6 months	Full calibration in ROTRIX; inspect cables and connectors; replace worn fasteners
After a crash or overload	Stop testing. Inspect the load cells and brackets, then run a full calibration before further use

NOTE Clean the sensors and the motor plate with a dry cloth after each session. Do not use solvents on the load cells or the circuit boards.

12. Troubleshooting

Symptom	Likely cause	What to do
Thrust reading drifts with the motor stopped	Load-cell cable under tension, or the load cells are still warming up	Re-route the cable with a loose service loop, wait 10 minutes after power-up, then zero again
Readings are noisy or jumpy	Loose bolts, an unbalanced propeller, or signal cables running beside power leads	Retighten the frame, balance the propeller, separate signal cables from the power leads
Torque reads a value with no load	Sensor block not square to the thrust axis, or the motor plate is touching the frame	Check alignment and clearances, then zero again
RPM shows zero or jumps	Sensor not aimed at the blade path, or too far away	Re-aim the sensor and move it closer, keeping it clear of the blade tips
DAQ unit not found by ROTRIX	USB cable, port or driver	Try another cable and port, check the driver, and restart ROTRIX
The stand creeps or rocks during a run	Clamps not anchored, or the floor is uneven	Stop, level the base and anchor every clamp

13. Support and contact

Contact	Details
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When contacting support, quote the document ID on the cover, the stand serial number, and the ROTRIX software version.

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